TWINPEACS: Towards a digital TWIN for Physical and socio-Economic Analysis of Coastal Systems using AI

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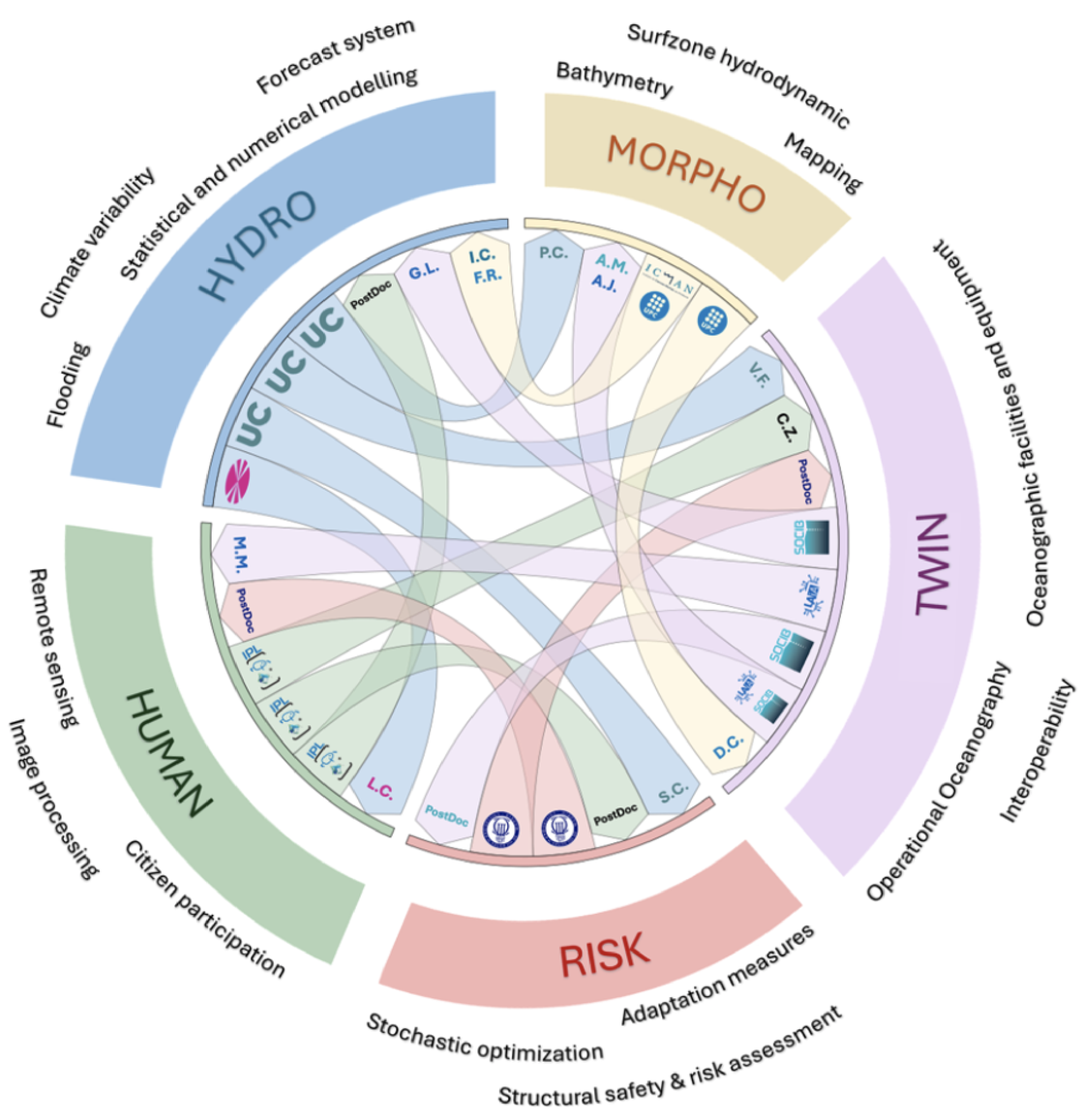
Motivation

- **The challenge**: Coastal cities are increasingly at risk from natural hazards due to climate change, rising sea levels and rapid population growth.
- **The need**: Advanced tools for reliable risk assessment and land use planning.
- **The TWINPEACS solution**: An interoperable, AI-enhanced DT framework designed for integrated coastal risk management.

Overview of the project

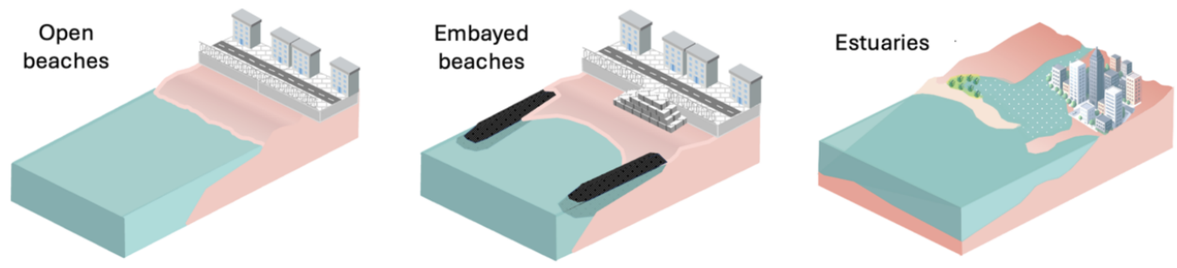
TWINPEACS is a **multi-disciplinary** initiative that transforms coastal risk assessment by incorporating AI into complex physical systems. To address compound hazards, such as erosion and flooding, we are developing a modular and scalable AI-driven DT framework. This framework will capture the non-linear interactions between **hydrodynamics**, **morphodynamics** and **socio-economic factors** that are driven by climate change and demographic shifts.

The project integrates five subprojects across nine institutions, bringing together different disciplinary perspectives from the fields of oceanography, civil engineering, computer science and social sciences.



TWINPEACS-HUMAN

- We build on a modular, **archetype**-based strategy that ensures the **scalability** and **transferability** of AI tools and workflows across diverse physical settings.



- Our **aim** is to generate dynamic, high-resolution exposure and vulnerability maps of Spanish coastal urban areas.

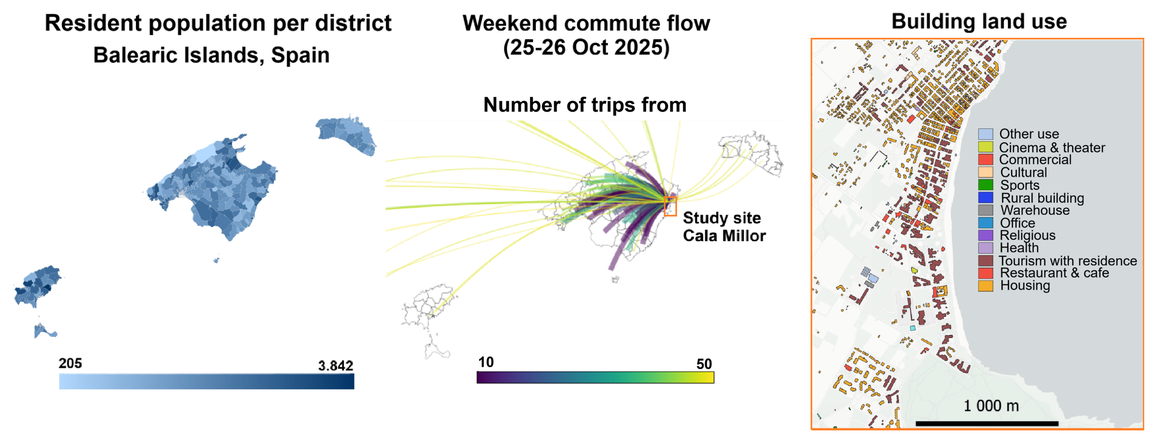
Specific objectives:

1. **Dynamic population disaggregation:** Develop ML models to estimate fine-scale population distribution across space and time [1].
2. **Socioeconomic vulnerability mapping:** Integrate socioeconomic, land use, and spatial data to derive composite vulnerability maps using statistical and AI techniques [2].
3. **Future scenario simulations:** Using DL to simulate how different urban planning trajectories will impact future risks.
4. **Integrated costal risk assessments:** Creating comprehensive risk maps that combine dynamic exposure and hazard data with uncertainty quantification.



Workflow

- **Combining multi-modal data**: socioeconomic statistics (census), mobility records (MITMA) with geospatial and remote sensing data (Sentinel, VIIRS, land use, and historical maps).



- **Training models** for population mapping, vulnerability assessment, and urban growth simulation.
- Integrating with the **European Digital Twin Ocean (EDITO)**—an EU virtual model translating complex ocean data into actionable knowledge. EDITO's high-performance computing will host our scalable DT prototypes, providing standardized, interoperable tools for real-world climate adaptation planning.

EDITO European Digital Twin Ocean

Consortium

UC Universidad de Cantabria · UNIVERSIDADE DA CORUÑA · UPC Universitat Politècnica de Catalunya BarcelonaTech · ICMAN Instituto de Ciencias Marinas de Andalucía CSIC · Institut de Ciències del Mar CSIC · UIB Universitat de les Illes Balears · LAiA Laboratori d'Aplicacions de la Intel·ligència Artificial · uc3m Universidad Carlos III de Madrid · SOCIB Balearic Islands Coastal Observing and Forecasting System · Universitat de València

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KEY REFERENCES

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MORE INFORMATION

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